

$^{33}\text{S}(^3\text{He},\text{d})$ [2014Pa44,1971Er03](#)

$^{33}\text{S}+1\text{p}\rightarrow^{34}\text{Cl}$ from $J^\pi=3/2^+$ ^{33}S ground state.

[2014Pa44](#): a 25-MeV ^3He beam was provided from the 14-MV MP tandem accelerator at the Maier-Leibnitz Laboratory, Garching, Germany. Targets were $\text{Ag}^2\text{ }^{33}\text{S}$ (20 $\mu\text{g}/\text{cm}^2$, enriched to 99.9% ^{33}S) deposited upon a foil of ^{12}C ; Ag^2S (20 $\mu\text{g}/\text{cm}^2$, natural sulphur) deposited upon a foil of ^{12}C ; a 50 $\mu\text{g}/\text{cm}^2$ Ag foil; a 7 $\mu\text{g}/\text{cm}^2$ ^{12}C foil; and ^{24}Mg (20 $\mu\text{g}/\text{cm}^2$, enriched to 99.92% ^{24}Mg) deposited upon a foil of ^{12}C . All ^{12}C foils were enriched to 99.99% and of thickness 7 $\mu\text{g}/\text{cm}^2$. Light reaction products were momentum-analyzed by the Q3D magnetic spectrograph and detected in the focal plane using a multiwire gas-filled proportional counter backed by a plastic scintillator at FWHM=12 keV. Measured $\sigma(E_d, \theta)$. Deduced levels, L-transfers, J, π , and spectroscopic factors from FRESCO-DWBA analysis of the measured $\sigma(\theta)$. Also deduced resonance strengths and the thermonuclear $^{33}\text{S}(\text{p}, \gamma)^{34}\text{Cl}$ reaction rate. Calculated $^{32}\text{S}/^{33}\text{S}$ ratios in the composition of nova ejecta via hydrodynamic nova simulations.

[1971Er03](#): a 14-MeV ^3He beam was provided from the Argonne tandem Van de Graaff accelerator. Targets were CdS on C backing, enriched to 83% in ^{33}S . Deuterons were detected using nuclear track emulsions in the focal plane of the Enge split-pole magnetic spectrograph with an energy resolution of 17 keV. Measured $\sigma(E_d, \theta)$. Deduced levels, L-transfers, J, π , and spectroscopic factors from JULIE-DWBA analysis of the measured $\sigma(\theta)$.

 ^{34}Cl Levels

E(level) [†]	L [#]	C ² S [#]	Comments
0	2	0.25	
145.5	2	1.42	
461.0 <i>12</i>	0+2	0.04+0.08	
665.1 <i>14</i>	0+2	0.07+0.50	
1229.4 <i>16</i>	0	0.12	
1885 <i>3</i>	0	0.02	
1923.3 <i>17</i>	2	0.04	
2157.5 <i>15</i>	2	0.66	
2579.1 <i>16</i>	0	0.03	
2608.5 <i>13</i>	2	0.08	
2720.7 <i>25</i>	1+3	0.04+0.39	
3128.0 <i>22</i>	2	0.11	
3333.0 <i>24</i>	2	0.09	
3382.7 <i>20</i>	2	0.31	
3544.9 <i>25</i>	1	0.18	
3631.4 <i>25</i>	3	1.10	
3773.0 <i>24</i>	1	0.05	
3983.4 <i>25</i>	3	0.44	
4076 <i>3</i>	3	0.52	
4146 <i>3</i>	(1+3)	0.08+0.24	
4353 <i>3</i>	1	0.14	
4418 <i>3</i>	1	0.11	
4461 <i>3</i>	1	0.09	
4516.3 <i>22</i>	1	0.13	
4607.9 <i>21</i>	1	0.11	
4638.7 <i>21</i>	1+3	0.010+0.017	L: other: 1 (1971Er03). C ² S: 0.010 2 for L=1, 0.017 3 for L=3 (2014Pa44). Other: 0.06 for L=1 (1971Er03).
4695 <i>2</i>			
4718 <i>2</i>			
4857 <i>2</i>	2,1+3	0.230 8	C ² S: for L=2. C ² S=0.0059 5 for L=1. C ² S=0.034 3 for L=3.
4941.9 [‡]	1	0.0060 7	
4957.3 [‡]	2	0.079 5	
4979 <i>2</i>	1,0	0.0058 4	C ² S: for L=1. C ² S=0.0095 7 for L=0.
4994 <i>2</i>	2	0.056 5	
5009 <i>2</i>	2,3	0.070 3	C ² S: for L=2, 0.020 1 for L=3.
5028 <i>2</i>	1	0.0010 1	
5171.6 [‡]	3,2	0.0159 7	C ² S: for L=3, 0.057 3 for L=2.

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$^{33}\text{S}(^3\text{He},\text{d})$ [2014Pa44](#), [1971Er03](#) (continued) ^{34}Cl Levels (continued)

E(level) [†]	L [#]	C ² S [#]	Comments
5246 2	3	0.112 3	
5338 2	1+3,2	0.0060+0.022	C ² S: 0.0060 5 for L=1, 0.022 2 for L=3, 0.200 8 for L=2.
5386.8 [‡]	3	0.0128 7	
5435 2	0	0.0110 6	
5453 2	0	0.0034 5	
5577 2	0	0.081 1	
5608 2	3	0.334 3	
5635.7 [‡]	1+3,2	0.0081+0.054	C ² S: 0.0081 7 for L=1, 0.054 2 for L=3, 0.247 4 for L=2.
5672.9 [‡]	0	0.0118 8	
5784 2	1	0.0010 1	
5805.9 [‡]	1+3	0.0028+0.016	C ² S: 0.0028 6 for L=1, 0.016 1 for L=3.
5852.8 [‡]	1+3	0.012+0.039	C ² S: 0.012 1 for L=1, 0.039 3 for L=3.
5897.1 [‡]	1	0.0088 7	

[†] For levels below 4.64 MeV, from [1971Er03](#); for levels above 4.64 MeV, from [2014Pa44](#), unless otherwise noted.

[‡] E(level) as given in [2014Pa44](#) without uncertainties, which observed these levels and adopted the E(level) values from the Adopted Levels of [2012Ni10](#) for calibration.

[#] From DWBA analysis of the measured $\sigma(\theta)$ in [1971Er03](#) (below 4.63 MeV) and [2014Pa44](#) (above 4.63 MeV). The spectroscopic factors C²S given by [1971Er03](#) are defined as $[(2J_f+1)/(2J_i+1)]C^2S$, where $C^2=1/2$ is the isospin coefficient, J_f and J_i , are the final and initial spins, respectively. $J_i=3/2$ for ^{33}S ground state.